

Samira Farhat

Beirut, Lebanon

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EDUCATION

Phoenicia University (PU) – Sarafand, Lebanon

October 2023 – Present

Bachelor of Science in Computer Science

Key Courses: Introduction to Programming, Intermediate Programming, Low-Level Programming, Algorithms and Data Structures, Computer Architecture, Database Systems, Mobile Programming, Computer Networks, Artificial Intelligence, Machine Learning, Software Engineering, Parallel and Distributed Computing

Expected Graduation Date: September 2026

American University of Beirut (AUB) – Beirut, Lebanon

October 2021 – June 2022

Completed Freshman Year in Computer Science (Transferred)

Al Mabarrat – Jouaiya, Lebanon

Awarded in June 2021

Lebanese Baccalaureate in Sociology and Economics

PROFESSIONAL EXPERIENCE

Simigo – Lebanon – Software Development Intern

August 2026 – Present

- Contributing to the development and maintenance of **kiosk applications and advertisement management dashboards** as part of a professional software development team.
- Developing and improving **frontend interfaces and dashboard functionality**, with a focus on usability, responsive UI implementation, and user-facing features.
- Working with **data handling and application logic** to support dashboard features and interactions within the system.
- Debugging and troubleshooting application issues across development and testing environments, identifying problems and implementing appropriate fixes.
- Reviewing and working with an existing production codebase to understand its architecture, functionality, and development patterns before implementing changes.
- Applying software engineering concepts in **frontend development, mobile application development, data management, and UI implementation** within a professional environment.
- Collaborating with team members to understand requirements, implement assigned features, and improve existing application functionality.

Freelance Work – Remotely

2024 – Present

- Helped clients and students with programming assignments and computer science coursework.
- Provided guidance in debugging code, understanding algorithms, and completing school projects.
- Developed simple websites and helped small clients with HTML, CSS, JavaScript, and basic back-end integration.
- Supported projects involving Python, Java, and databases (MySQL), including fixing bugs and improving functionality.

DOOBOT (USA) – Remotely

July 2024 – December 2024

Web Development Department

Intern

- Assisted in designing and building a responsive website, enhancing front-end functionality with HTML, CSS, JavaScript.
- Participated in team meetings, gaining experience in collaboration, problem-solving and optimization.
- Conducted debugging and resolved performance issues to improve website efficiency.

Fresh Line – Tyre, Lebanon

March 2024 – August 2024

Sales Department

Sales Associate

- Operated POS (Point-of-Sale) systems using SaaS technology to process transactions efficiently.
- Delivered exceptional customer service and maintained organized product displays.
- Worked with the team to exceed sales targets and improve the customer experience.

PROJECTS

X-Pensa – Full Stack Application

August 2026

GitHub: <https://github.com/samira-farhat/expense-tracker>

- Collaborated as part of a development team to design and develop X-Pensa, a full-stack personal finance application for managing income, expenses, transaction history, and financial reports.
- Built the backend using **Node.js, Express, and TypeScript**, implementing RESTful APIs for authentication, users, transactions, balances, and financial reports.
- Designed and managed a **MySQL** database for users and transactions, including user-specific transaction relationships and data retrieval.
- Implemented secure authentication using **JWT**, password hashing with **bcrypt**, and protected API routes.
- Developed the responsive web frontend using React and TypeScript, integrating backend APIs with Axios and implementing a mobile-friendly dashboard layout.
- Built transaction management features including adding, editing, deleting, searching, filtering, and date-based transaction restrictions.
- Implemented financial analytics including monthly reports, yearly expense charts, category expense distribution, income/expense summaries, and balance calculations.
- Developed a **Flutter mobile frontend** connected to the same backend API, extending the application to Android devices with persistent authentication using secure local storage.
- Implemented input validation across the frontend and backend, including transaction amount and category validation.

GaitAnalytica – Senior Project – PU

March 2026 – July 2026

GitHub: <https://github.com/samira-farhat/gaitanalytica>

- Developed an AI-powered smartphone-based gait analysis system for rehabilitation monitoring by extracting biomechanical insights from walking videos.
- Implemented computer vision pipelines using **Python, OpenCV, and MediaPipe Pose** for human pose estimation, landmark tracking, and gait cycle analysis.
- Extracted and analyzed biomechanical gait metrics including **knee range of motion, step length, cadence, stride timing, symmetry, and walking efficiency** using movement-based calculations.
- Built a full-stack application with **Django REST Framework backend** and mobile frontend integration for video upload, gait processing, visualization, and personalized analysis reports.
- Integrated **Google Gemini AI** for automated interpretation of gait metrics and generated user-friendly biomechanical feedback from analyzed sessions.
- Designed asynchronous background processing using **django-q2 task queues** to handle computationally intensive AI analysis and report generation without blocking user requests.
- Implemented a consultant communication workflow allowing users to securely share gait reports and historical session data with healthcare professionals.

ML Classification Analysis for Breast Cancer Detection– PU

June 2026

GitHub: <https://github.com/samira-farhat/breast-cancer-ml-project>

- Developed a machine learning classification pipeline to predict benign and malignant breast cancer cases using the Wisconsin Breast Cancer Dataset.
- Implemented and compared multiple supervised learning algorithms, including Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN).
- Performed data preprocessing, feature scaling, model training, and hyperparameter optimization using Python and Scikit-learn.
- Evaluated model performance using accuracy, precision, recall, F1-score, confusion matrices, ROC curves, and AUC analysis.
- Analyzed and compared model results, achieving 98.2% accuracy with SVM as the best-performing classification model.

Car Rental Management System – PU

January 2026

GitHub: <https://github.com/samira-farhat/car-rental>

- Collaborated in a four-member team to design and develop a comprehensive mobile-based car rental management system for a local business (*Drive with Khachab*).
- Developed the mobile frontend using **Flutter**, implementing a secure and user-friendly interface for browsing, searching, and renting vehicles.
- Designed a centralized **MySQL** database to manage user roles (Admin, Manager, Customer), car inventory, and real-time availability.
- Integrated a **Python-based backend API** to handle secure communication, user authentication (using hashed credentials), and payment processing.
- Built advanced features including an automated document upload system for driver's licenses, a reservation engine with a 7-day limit, and automated report generation for business income and usage.
- Conducted full requirement analysis and process modeling, creating **ER Diagrams** and **Data Flow Diagrams (DFD)** to optimize the business's digital transformation.

Malaria Cell Image Classification Using CNN – PU

January 2026

GitHub: <https://github.com/samira-farhat/malaria-cell-classification-cnn>

- Developed a Convolutional Neural Network (CNN) to classify malaria-infected and uninfected blood cell images.
- Preprocessed 27,558 microscopy images through resizing (64×64), normalization, and label encoding.
- Designed a deep CNN architecture with convolutional, pooling, dropout, and dense layers using TensorFlow/Keras.
- Trained and optimized the model on Google Colab with GPU acceleration.
- Evaluated performance using accuracy, precision, recall, F1-score, and confusion matrix.
- Achieved approximately **95% test accuracy**, demonstrating strong generalization and minimal overfitting.
- Visualized training behavior via accuracy and loss curves.

Mobile Bookstore App – PU

March 2025 – June 2025

GitHub: <https://github.com/samira-farhat/bookstore-app>

- Developed a mobile e-commerce app for a bookstore as part of the Mobile Programming course.
- Integrated an API for user authentication (using Reqres).
- Implemented product browsing, wish list, cart management, and order tracking as features.
- Utilized Flutter with state management and local storage to maintain user sessions and cart data.
- Designed an intuitive, responsive UI focused on a smooth user experience.

Database System for Hera Flower Shop – PU

March 2025 – May 2025

GitHub: <https://github.com/samira-farhat/hera-flowershop-db>

- Designed and implemented a complete database system for a real flower shop business.
- Gathered requirements, did ER modeling, normalization, and wrote SQL queries.
- Used MySQL for database creation, query handling, and reports generation as tools.
- Built a user-friendly, multi-window GUI using Python's Tkinter to manage items, customers, suppliers, and orders.
- Integrated advanced business logic, including stock level adjustments, loyalty points system, intelligent order cancellation handling, and dynamic total price calculation based on item discounts and quantities.

Nexus Student Hub – PU

November 2024 – February 2025

GitHub: <https://github.com/samira-farhat/nexus-student-hub>

- Developed a student hub website as part of a web development course, providing academic resources and opportunities for students.
- Implemented using HTML, CSS, and JavaScript with user-friendly navigation.
- Focused the project solely on front-end development.

CERTIFICATIONS

Python Essentials 1 & 2 – Cisco Networking Academy

August 2023

Summer Camp Participant – Beirut Digital District (BDD)

July 2019

Earned a certificate in the following areas:

- **Woodworking & Electrical Engineering:** Built a moving wooden model with electrical components.
- **Game Development:** Created a 2D Mario-style game with gameplay mechanics.
- **Creative Writing:** Developed writing and storytelling skills.
- **Photography and Videography:** Created a short film, including editing and post-production.

EXTRACURRICULAR ACTIVITIES

Computer Science Society – PU

October 2023 - Present

Member

- Participating as an active member in events and discussions to foster technical growth.

SKILLS AND INTERESTS

Linguistic Skills: Fluent in English and Arabic. Basic in French and Spanish.

Technical Skills:

- **Programming Languages:** Java, Python, C++, C, C#, JavaScript, TypeScript, SQL.
- **Machine Learning & AI:** Scikit-learn, TensorFlow/Keras, Convolutional Neural Networks (CNNs), Model Training & Evaluation, Data Preprocessing, Feature Scaling, Hyperparameter Tuning.
- **Computer Vision:** OpenCV, MediaPipe, Image Processing, Human Pose Detection.
- **Parallel & Distributed Computing:** Parallel Programming Concepts, Performance Optimization, Multithreading/Parallel Processing, OpenMP, OpenCL, MPI, and Big Data (Hadoop and PySpark).
- **Web & Backend Development:** React, Node.js, Express.js, TypeScript, REST APIs, Axios, HTML, CSS, Responsive UI Development, JWT Authentication, bcrypt
- **Mobile Development:** Flutter, Dart, Android Development, Local Storage, API Integration
- **Database Tools:** MySQL, Microsoft Access, Relational Modeling, Normalization, ER Diagrams, SQL Queries.

- **Data & Development Tools:** Git/GitHub, Google Colab, Jupyter Notebook, NumPy, Pandas, Matplotlib.
- **AI & Digital Tools:** Skilled in leveraging AI tools like ChatGPT to enhance learning, debug efficiently, explore new technologies, plan projects, and improve overall productivity.
- **Game Development:** Basic exposure to Unity with C# scripting (self-taught).
- **Productivity Tools:** Microsoft Office (Word, Excel, PowerPoint, Access).

Hobbies: Reading, Basketball, Baking, Writing, Playing Ping Pong, Playing Padel.